

Computational Phenomenology*

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Abstract: This paper provides a conceptual and historical analysis of “computational phenomenology,” covering its various contemporary forms and tracing its historical development back to Husserl’s early work on algorithms and mechanical computation. The virtues (and vices) of computational modeling are enumerated and used to explain why computational models are valuable (and dangerous) in phenomenology. The paper ultimately defends a piecemeal approach to computational phenomenology, whereby models at a range of temporal and spatial scales are used to understand the many domains and themes of phenomenology, from low-level perception to skilled behavior to abstract thought. The hope is that an increasingly coherent story about phenomenology and its neural realization emerges from this patchwork of modeling efforts.

Keywords: computation, naturalized phenomenology, Husserl, IIT, predictive processing, active inference, computational theory of mind, Heidegger.

Introduction

The history of phenomenology intersects with the history of computation in surprisingly many ways. Husserl actively contributed to discussions surrounding the foundations of logic and developed a theory of algorithmic and computational procedures in his early work. On his account, algorithmic processes are essential to the advance of human knowledge but become dangerous when followed blindly. In the 1960s researchers began to take note of various overlaps between Husserl’s work, artificial intelligence (Ken Archer, this volume), and early cognitive science (Shaun Gallagher and Vittorio Gallese, this volume). In the 1990s, consciousness studies emerged and phenomenology began to be linked to a broader array of computational approaches in cognitive science, including neural networks and dynamical systems theory. Today there are several prominent theories of consciousness—especially integrated information theory and active inference approaches—that make explicit use of computational concepts in explaining consciousness, and it is the latter, namely active inference, that has popularized the term “computational phenomenology.”

The time has come for a pause and reflection. How do these various accounts relate to each other? Of what value is computation in phenomenology, and what are the dangers?

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This chapter argues that a coherent story can emerge, which can be organized around the virtues and vices of computational models. An especially promising approach, it is argued, is one that focuses on specific cases—the phenomenology of time, visual experience, skill, and related domains—and in each case formulates computational models that allow for precise formulations of phenomenological claims. The goal is to facilitate the gradual emergence of a comprehensive account of the neural basis of consciousness, one that encompasses multiple scales and a plurality of theoretical frameworks.

1. Approaches to Computational Phenomenology

There is no consensus about how to define “computational phenomenology.” On one conception, it denotes efforts to *account* for phenomenology using computational models. Such views can be cashed out in various ways, for example as efforts to explain or model consciousness. On some of these views, consciousness itself is taken to be a form of computation or emergent from a computational process. On a second conception, the term denotes any effort to use computational tools in phenomenology, whether or not the ultimate goal is to “account” for consciousness.

Further complicating the terminological issue is the fact that “phenomenology” is polysemous. On the one hand, there is “phenomenology” as an approach to fundamental ontology, what Husserl called “transcendental phenomenology” and studied using such methods as *epoche* (“bracketing”) and eidetic variation. Phenomenology in this sense is practiced independently of external, for example, naturalistic considerations. On the other hand, there is “phenomenology” in the sense of everyday introspection or reflection on conscious states.

In this paper I will focus on the second usage in each of these cases: that is, on any use of computational tools in phenomenology, where “phenomenology” is understood to encompass any effort to describe conscious states or processes, transcendental or otherwise. The virtue of this choice is that it is inclusive of all usages of “computational phenomenology” I am aware of.¹

Computational models have many virtues, which can be usefully contrasted with the virtues of mathematical models. Computer models can simulate processes and make predictions that would be impossible for a human to produce on paper in a reasonable amount of time. Call the new insights made possible by computers “computationally mediated insights.” Computer simulations also allow for fast iterative refinement. They can be quickly tweaked and refined, as opposed to a mathematical line of reasoning which must be slowly checked by a human. These features support social virtues. Once we have a running model, others can immediately use it, often simply by cloning a source repository or copying a notebook and executing it themselves. In verbally communicating mathematical reasoning, by contrast, ambiguities can interfere and each interlocutor must have enough mathematical training to understand the problem. With computer

simulations, many variants of an idea can be developed relatively quickly and collaboratively, and sometimes even competitively (as in hackathons or coding competitions).

However computational models can also be problematic. The main issue from a phenomenological standpoint is that they introduce non-phenomenological evidence into phenomenological inquiry. The whole point of the phenomenological reduction in Husserl is to preclude non-intuitive sources of justification from transcendental phenomenology, and to only describe things as they are given. Even in everyday phenomenology this can be a problem, insofar as reliance on complex models with many latent variables can potentially mislead us and produce distorted descriptions or accounts of experience. The problem is exacerbated when computational models are especially complex, with millions or billions of hidden variables (as in a large language model). In these cases, the model becomes a kind of black box, and we must then take it on faith that something inside of it sheds light on phenomenology. Call this the “vice of remoteness from intuition,” which is closely related to what Husserl calls “inauthentic” thinking. As we will see, Husserl allows for such concepts and theories to be used, as long as we have a clear understanding of how they work and can independently check their results against what he calls “originary intuitions.”

So we have a project: using computational models to help describe and understand conscious processes. How, in practice, does this work? Does one start with phenomenology or computational models or use both at once? Here are three broad approaches, which are inspired by similar approaches in naturalized phenomenology.

One approach is “phenomenology-first”. This approach starts with phenomenology and then develops computational models to make sense of it. That is, we only use computational models *after* we have gotten the phenomenology straight. This is related to “frontloaded” phenomenology.² As we will see, integrated information theory (IIT) is a phenomenology-first approach, and uses this term explicitly.

A second approach is “computation-first”. This approach starts with a computational model that does something intelligent, like a Large Language Model (LLM), and ask if it sheds light on some topic in phenomenology. This can also be called “backloaded” phenomenology.

A third approach can be referred to as “convergent phenomenology”. This approach pursues both phenomenology and computational modeling and allows them to mutually influence one another until they converge on a common approach or settle into a kind of reflective equilibrium.³

Arguably, front and backloaded approaches are really just variants on the convergent approach, since anyone doing this kind of work will have both phenomenology and computation in mind. Still, in practice, front-loaded approaches treat phenomenology as the ultimate evidential standard, and computation-first approaches begin with a model

(that might have been built without any thought about phenomenology) and then try to understand it in terms of phenomenology.

2 Husserl's Theory of Computation

Before discussing computational phenomenology as practiced today, it will help to review Husserl's own account of computation. This is not the same thing as computational phenomenology: it is an account of computation in terms of phenomenology rather than an account of phenomenology in terms of computation. That is, it is a phenomenology of computation. Still, it sheds light on several aspects of the later discussions and develops important points about both the promise and danger of computational processes.

In the *Philosophy of Arithmetic* (1891), Husserl studied how numbers and simple arithmetic operations are given to the mind in cognitive acts.⁴ For example, in seeing a group of apples as three apples, they must each be seen as a "something" and then be combined in an act of collective combination, which requires sequentially enumerating them.⁵ This is the basis of the authentic concept of number, where an "authentic concept" is one "acquired through the immediate presence to our mind, in an adequate or full intuition, of the essence corresponding to (but not identical with) the concept."⁶ In humans, only about two through seven are given authentically. Even zero and one are not authentic number concepts for Husserl, because "they incorporate no 'collective combination,' and in the case of zero not even any 'something.'"⁷ As he says, "The designation of zero and one as 'numbers' represents a *transference* of that term to concepts of a different kind"⁸, though "motives of the greatest importance, speak in favor of this transference."⁹

Husserl thus turns to inauthentic number concepts, which are the basis of most number concepts, and ultimately the basis of all mathematical reasoning and calculation. They are essential to reasoning. Without them, we would not have the vast majority of mathematics, or even arithmetic: "the whole of arithmetic is, as we shall see, nothing other than a sum of artificial devices for overcoming the essential imperfections of our intellect."¹⁰ Inauthentic concepts use symbolic surrogates to refer to numbers and are governed by rules of replacement. For example, zero occurs when the operation of subtraction is applied to the signs for "3" and "3." Husserl analyzes various extensions of the number system in this way (negative, rational, real, and imaginary numbers), and in the 1890s developed a theory of algorithms. He defines "an arithmetical algorithm" to be "a system of formal rules by means of which we can solve problems concerning numbers by purely mechanical operations, i.e. from known numbers and number relations unknown ones can be discovered."¹¹

Husserl's account of algorithms involves a mapping between abstract concepts and physical signs, and the introduction of rules governing the signs. This allows us to offload the thinking procedure to an algorithm:

In order to solve a problem, to derive a proposition, we need not think about the concepts themselves, but we can make do with a certain rule-based process to connect signs with signs, replace sign-complexes with sign-complexes, etc. Calculating is an operation with signs and not with the concepts themselves, and the result of the calculation is again something purely signitive, a certain combination of signs on paper.¹²

Husserl also understood that algorithmic operations could be carried out by mechanical means and refers to calculation devices throughout his works. Moreover, he recognizes the virtue of computationally mediated insight, seeing calculation devices as tools that allow reasoning to extend far beyond what would be possible using authentic thought alone.

However, there is a danger here. If we only operate blindly with symbols, or algorithms, or computing machines, we can never be confident in the results we obtain. We must justify algorithms and mechanical procedures in terms of authentically given rules and see for ourselves that they work properly when extended beyond what can be immediately given.¹³ For example, we know authentically what it means to add a small number to another small number for authentically given pairs of numbers using a mechanical device that simply increments a counter, and we can thus see how the system could produce valid results for additions that could *not* be authentically given. In a similar way, we can see how a system of gears could work to add and subtract small numbers, and by confirming its operation on small numbers come to trust it when it performs operations on large numbers. In this way, we can outsource our thinking to blind procedures, but ideally only to those we have checked in an appropriate way.

So the key, for Husserl, is to periodically confirm (in authentic thought) how the inauthentic thought processes and calculation devices we use actually function. When we fail to do that, when we simply take the inauthentic processes for granted, outsourcing our thought to them and getting lost in a “game with symbols,” we risk the kind of “crisis” described in his late work. As Ierna says:

There is a very serious risk ... when we forget to “trust, but verify” the technologies we use to do science and conduct our daily lives. Husserl underscores this problem in the *Crisis* when he speaks of the technicization of science. The risk is not so much that we miss our targets in doing science, but that our aims and values themselves are perverted... For Husserl this “technicization” is linked directly to the use of mathematical symbolism... We translate concepts into signs, play with the signs, and then translate the signs into concepts. In that process, when extended to science as a whole, we risk forgetting our original aims and values, for the sake of

which the whole symbolic technique was invented and implemented in mechanical devices in the first place.¹⁴

3. Phenomenology and the Computational Theory of Mind

Building on the observations of the last section, numerous writers have noted that a kind of computational theory is at work in Husserl, some version of what is often called the computational theory of mind, or CTM. CTM emerged from the work of Noam Chomsky, Jerry Fodor, Zenon Pylyshyn, and others in the 1960s and 1970s, as a way to formalize early cognitive science into a scientific research program. The theory posits that mental processes operate on mental representations (in this regard, it overlaps with what is sometimes called the representational theory of mind, or RTM) in accordance with formal rules and treats the syntactic rules of language as a useful model of the rules governing mental processing. Early work in artificial intelligence often involved the use of computer programs, which used rules and knowledge bases to solve problems, process language, and play games like chess. It was argued that these were cognitive simulations, which process information in something like the way humans do. In some cases, humans would report on their own mental states when solving problems, and these “protocol statements” (which were really just rough-hewn phenomenological reports) were compared to the traces of computer programs solving the same problems, to provide evidence that humans and computers solve problems in the same way.

Numerous authors from the 1970s and 1980s commented on linkages between this emerging research program and phenomenology, especially Husserlian phenomenology. Multiple authors noted that Husserl held some version of a representational theory of mind.¹⁵ Intentional states are representational states, and they are central to Husserl’s account—Husserl’s teacher Brentano famously described intentionality as the “mark of the mental.” An overwhelming part of our phenomenology is representational in character. In everyday experience, we are oriented towards various types of objects in perception, thought, and language. It was also noted that Husserl, like Chomsky, describes rules for how these representations can be combined and recombined, and focuses on the case of language as paradigmatic.¹⁶ Moreover, Husserl describes rules for organizing perceptual information and expectations in perception, using ideas similar to those used by Marvin Minsky and other early artificial intelligence researchers.¹⁷ The method of cognitive simulation, where program traces are compared to human thought processes, was seen to resonate with early Husserl’s analysis of authentic and inauthentic thought (as discussed in the last section), where a person solves a problem, then tries to understand and justify how they did it.¹⁸

Harlan, whose main paper on these topics is titled “Towards a computational phenomenology,” nicely summarizes this early work:

At first glance, largely because the terminologies are so radically different, it appears that the descriptions of mind that have resulted in the two traditions share nothing in common: whereas the phenomenologist speaks of apprehension, motivation, active and passive genesis, sedimentation and habitualities, the cognitive scientist speaks of symbols and expressions, data bases, control structures and searches.¹⁹

However, Harlan continues, in both cases, the mind is thought of as a kind of structured field.

Stored in the system are perceptual patterns characteristic of object types, information about the values of objects (e.g., practical uses), the roles objects play in various contexts, etc. The information stored in the system is the product of experience (Husserl, against Kant, sided with the empiricists here). In particular the information is the product of the way the system organized input from the environment.²⁰

In recent years, the link between the computational theory of mind and Husserl's phenomenology has been systematically developed by Jesse Lopes.²¹ Lopes emphasizes Husserl's early writings on signs and algorithms. A key passage for Lopes is the "thought machine" passage in the *Logical Investigations*. In this passage, Husserl begins by noting that calculators are mechanisms regulated by physical laws, which act in accordance with the laws of arithmetic:

The arrangement and connection of the figures which spring forth is regulated by natural laws which accord with the demands of the arithmetical propositions which fix their meanings.²²

Husserl goes on to say that a calculator is "no thought-machine, it understands neither itself nor the meaning of its performances." On the other hand, humans *are* thought machines, because our computations are also based on physical processes in the brain that act in an appropriate way, but *we* understand the meaning of those performances when we compute authentically (see Section 2). Husserl then notes that this ability to evaluate our own computations must be based on the operation of a separate system, perhaps another thought machine, that functions to evaluate the first one.

... our own thought-machine might very well function similarly, except that the real course of one kind of thought would always have to be recognized as correct by the insight brought forward in another. This latter thinking could be the product of the

same or other thought-machines, but ideal evaluation and causal explanation would none the less remain disparate.²³

This account bears a striking resemblance to the CTM, which also posits that human thought is based on causal laws which act in accordance with the ideal laws of a computational system, and in which the ability to produce appropriate behaviors is subject to evaluation by a separate system. Thus, Lopes argues, Husserl was a proto-computationalist, who discovered the CTM over a hundred years ago.

Other approaches to computational cognitive science have also been linked to Husserl and phenomenology, in particular connectionist, dynamical systems, and Bayesian approaches.²⁴ These approaches share CTM's emphasis on computation but make use of different kinds of computational models and explanatory schemes. In fact, the debates between these approaches and CTM have themselves been linked to debates within phenomenology between Husserl and Heidegger.²⁵ Some of the modeling efforts resulting from these other approaches to computational cognitive science are discussed in relation to phenomenology in Section 6.

4. Integrated Information Theory (IIT)

IIT bills itself as “phenomenology first” (in the same sense as is used in Section 1) and formalizes phenomenology in terms of a series of axioms.²⁶ These describe commitments to the existence of consciousness (the axiom of existence), the unity of consciousness (the axiom of integration), and the structure of consciousness, in the sense of parts and relations within conscious states (the axiom of compositionality). These phenomenological axioms were formulated independently of consideration of philosophical phenomenology,²⁷ though some have begun to consider the links between the two.²⁸

Each axiom is linked to a computationally defined mechanism via a corresponding postulate. For example, the postulate of information quantifies the degree to which current states rule out future and past states, and the postulate of integration quantifies the degree to which the elements of a system interact with each other, doing something more than the sum of their parts. For example, an array of disconnected sensor nodes carries little information (it does not rule out many past or future states) and is not integrated (partitioning does not change how it works), while a densely connected neural network is highly informative and integrated, since its elements work together in a way that is highly constrained, ruling out many past and future patterns of activity (information), and since partitioning changes how it behaves (integration). Integration and information are key components of a single measure of integrated information called “Phi.”

One of the goals of IIT is to be able to use Phi to determine whether a system is conscious based on observation of its internal structure and behavior relative to systematic interventions. This aspect of the project has yielded impressive results, motivating a

diagnostic measure for consciousness called the perturbational complexity index (PCI), which can be thought of as a proxy for Φ . PCI can be used to determine, quite accurately, whether someone is awake, dreaming, in a minimally conscious state, or a vegetative state.²⁹ The measure works by perturbing a system and measuring the response. When the response is suitably complex, it tends to be associated with consciousness. This work is notable for yielding working results that can be intuitively linked to phenomenology. Think of throwing a rock in a pond and seeing not a simple ripple. Now compare that with perturbing a system and seeing a self-organized pattern that unfolds in a highly patterned way. Being awake and normally conscious involves a rich tapestry of reactions to external events, and it makes sense that our waking phenomenology would correspond to this kind of informative and integrated activity.

Another goal of IIT is to determine what specific conscious contents occur in a system. That is, to be able to predict phenomenological contents based on the physical state of a system. Current research is developing IIT's account of time consciousness and spatial consciousness along these lines.³⁰

IIT manifests both the virtues and vices of computational phenomenology. Executable code to run the algorithms associated with IIT is available in public software repositories, and there are multiple tutorials and code-alongs online, as well as regular workshops.³¹ This allows for communal coding in what has become a large and highly active community of researchers.

However, remoteness from intuition is a concern. Φ is extremely complicated, so much so that it can't even be computed for most real-world datasets. That issue has been mitigated to some extent by the introduction of approximate measures like PCI, which work well for evaluating the presence of consciousness and for determining levels of consciousness. But the situation is not as good for content, at least not yet. Here, the vice of remoteness is more pronounced in that case, and no mitigating approximations or simplifications are currently available. As one participant said at a recent consciousness conference, referring to IIT's account of temporal experience: "I look at all the graphs and equations, and I just don't see time consciousness."

The issue is not insuperable. Any area of science can be difficult, and this is the frontier of IIT research, where systematic investigation has just begun. IIT-inspired work has a track record of success (for example, classifying levels of consciousness using PCI), and the social virtues of computational work are on full display as theorists try to simplify and streamline the theory, make it predictive, and link it more effectively to phenomenology.

5. Active Inference

The term "computational phenomenology," as it has been used in the recent literature, is most closely associated with "active inference," which is a covering term for a

range of views, including predictive processing and free-energy-based approaches to mind. These approaches are based on a few simple ideas, which extend naturally to phenomenology.³²

The origins of these approaches are in predictive coding accounts of the brain, according to which top-down prediction signals in the cortex are compared with external sensory information, resulting in error signals, which indicate when the brain's predictions are disconfirmed or surprising. The error signals are what get passed up through the visual system. For example, in a visual scene, the edges are surprising relative to patches of constant color, so edge information is what gets passed on to higher levels of the visual system. If I see a familiar building but there is an unfamiliar person at the door, the person is surprising relative to my normal expectations, and so that is the primary signal passed upstream. The broader idea is that internal statistical models or “generative models” (generative because they generate predictions) determine most of what we see, with inputs correcting and updating those models where necessary.

Predictive processing accounts generalize this conception of the visual system to the brain as a whole, treating it as a hierarchical collection of statistical machines generating top-down predictions and comparing those predictions with incoming signals, so that error is the main information passed upstream from one internal model to the next. The models are usually trained using Bayesian methods and are often described using network diagrams where boxes and circles correspond to the parameters of statistical models. Predictive processing is a kind of umbrella term for a family of theories with similar commitments, though there are important differences between them.³³

These accounts encompass perception, action, and cognition. In action, motor commands are reconstrued as predictions about sensory experience. When I raise a coffee cup to my mouth, I am predicting how my arms and the coffee will feel, and the primary information coming back from the motor system is prediction error, indicating where things were not quite as expected. On this account, “action is modeled as a kind of self-fulfilling prophecy: agents predict what state they will be in upon acting, and then generate evidence for this prediction by actually acting in the environment.”³⁴ Higher cognition can be understood as prediction operating on increasingly abstract internal models. For example, a tendency to enjoy Russian cinema is a parameter of a policy model which predicts a state of enjoyment upon watching such a movie, which in turn predicts specific actions which are carried out to fulfill those predictions.

This theory can be derived from an abstract principle known as the free energy principle (FEP), inspired by ideas in physics, according to which cognitive systems behave in ways that tend to minimize surprise over time, or, more specifically, surprisal, the negative log probability of sensory outcomes under an internal model. Updating such a model directly using Bayes' rule introduces the problem of having to compute integrals over internal model states, which is intractable in most realistic situations. As a result, “variational” methods have been introduced, which approximate Bayesian updating in a

tractable way, and still have the result that sensory outcomes tend to become less surprising over time.

So the overall account involves a formal foundation, a neural implementation, and can be understood in terms of hierarchically organized statistical models. These models can be implemented in simulations of specific neural and cognitive phenomena, manifesting all the virtues of computational models described in Section 1, including computationally mediated insight, iterative refinement, and communal integration. The approach is popular, and thousands of articles have been published using these ideas to explain virtually all the primary phenomena of cognitive science, including perception, motor control, decision-making, sense of self, social cognition, attention, and mental imagery.³⁵ Many of these studies are accompanied by running models available on public repositories, and, as with IIT, there is an active community of researchers collaborating on modeling projects. The computational models used in active inference are simpler than those in IIT, so that the vice of remoteness is arguably less prominent, making it easier to link the structure of these models to phenomenology than is the case in IIT.

Many features of active inference resonate with phenomenological themes, given the emphasis Husserl and others place on expectation (recall from Section 3 that parallels between Husserl and “expectation-driven theories” are also noted by Dreyfus, Harlan, and others). Husserl emphasizes that experience is organized around expectations and prior associations, and claims that, in moment-to-moment experience, “protentions” are compared with what actually occurs, resulting in experiences of “fulfillment” and “frustration.” Recognizing parallels like these, in 2022, Maxwell Ramstead began hosting a series of Friday meetings on “computational phenomenology,” and that same year, published a multi-authored paper titled “From generative models to generative passages: A computational approach to (neuro)phenomenology.” It opens with the following statement:

This paper presents a version of neurophenomenology based on generative modelling techniques developed in computational neuroscience and biology. Our approach can be described as computational phenomenology because it applies methods originally developed in computational modelling to provide a formal model of the descriptions of lived experience in the phenomenological tradition of philosophy (e.g., the work of Edmund Husserl, Maurice Merleau-Ponty, etc.).³⁶

This is a computation-first account, insofar as the computational theory came first, and was later applied to phenomenology. However, it is being developed in a highly convergent way. The authors “propose a new version of neurophenomenology as ‘generative passages,’”³⁷ which seeks “bona fide mutual epistemic accountability—and virtuous circularity—in the generation of scientific explanations”³⁸ The term is a play on generative models, which are both the key computational structure in active inference but

also what provides for “back-and-forth passage” between first and third-person accounts of the mind.

Work in this emerging tradition has connected many aspects of predictive processing, active inference, and free energy models to specific features of phenomenology, particularly in Husserl. For example, predictions map to protentions, errors map to surprise or frustration, sensory inputs map to hyletic data which constrain interpretation of perceptual scenes, model inferences map to constitutive phenomenology, and so on.³⁹ Details of some of these accounts are given in the next section.

6. Piecemeal computational phenomenology

The cases discussed so far all pursue “computational phenomenology” in the context of a larger theoretical project (CTM, IIT, or active inference). But the idea can also be applied in a more modular or piecemeal way, where specific aspects of phenomenology are studied using computational models. The idea is to pursue neurophenomenology at many temporal and spatial scales.⁴⁰ One study might consider the phenomenology of vision in relation to processes occurring over hundreds of milliseconds, another, the phenomenology of perceptual processes occurring over tens of seconds, and another study might consider processes of meaning “sedimentation” unfolding over days or months. Other studies could cross scales, for example a study of how narrative self-understanding (a process that unfolds over many *years*) influences momentary perception. This section considers a series of examples starting at smaller scales and working upward. Some are phenomenology-first, some are computation-first, and some more convergent.

We can start with a relatively low-level case: the detection of edges in visual scenes. This is a case where Husserl describes the process in about as much detail as he can using purely phenomenological resources, and where computer scientists and neuroscientists independently described it in a similar, but more detailed, way. Husserl gives the following phenomenological description of the process: “A ‘gradual’ transition through a gradually changing qualitative gradation yields a border only if the transition first goes very slowly and then proceeds very quickly, and then very slowly again.”⁴¹ Decades after Husserl said that, edge detection became an important area of research within computer vision. Many edge-detection algorithms use some variant of Husserl’s rule, looking for sudden changes in image brightness. Moreover, it is now long-established that orientation-selective cells in the visual cortex have a similar response profile to edge-detectors using that algorithm. So we have a nice convergence between phenomenology, computational principles, and neural implementation.⁴²

The computational precision of this research does more than facilitate convergence; it also suggests extensions and refinements. There are exceptions to Husserl’s “slow-fast-slow” rule, for example, complex textures with many such transitions that are simply perceived as textures. Local edge-detection is typically supplemented by global

information, and there is a theoretical edifice that distinguishes edges from boundaries and contours. All of this is suggestive of a computational phenomenology of low-level visual awareness that is more detailed and elaborated than would be possible solely on the basis of phenomenological resources.

Computational phenomenology can also be used to make progress on controversies that are difficult to resolve on purely phenomenological grounds. Consider the case of hyletic data, a kind of sensory data which Husserl claims is core to perceptual experience. According to Aron Gurwitsch, Husserl is making a mistake in positing such data, and is in fact violating his own method of phenomenological reduction.⁴³ The idea that we experience sensations is motivated by consideration of how the brain works, rather than being something we encounter when considering things just as they are given, independently of naturalistic or other external considerations. According to Gurwitsch we never directly experience a distinction between sensory inputs and conceptual interpretation. Rather, what we experience are fully-formed *Gestalts*, whole perceptual units which contain no traces of pointillistic sensory data. I always found this to be persuasive but still thought there was some sense in which we were aware of sensations.

Dagfinn Føllesdal provides a way out, and does so in a way which suggests links to computational models of the brain. He argues that hyletic data impose *boundary conditions* on the interpretation of a stimulus, that they function as something “ego foreign” that enters into our acts and limits what we can experience.”⁴⁴ They are not directly present in experience, but we can infer their presence. I can see the apple in many ways, but the space of things I could see is constrained by some kind of brute presence beyond me: I could not see the apple as a dragon or soccer ball. That sense of something ego-foreign is given. Active inference provides a nice way of explaining hyletic data which supports Føllesdal’s reading. Generative models generate top-down predictions, and these are combined with sensory-inputs to produce our overall perception. Sensory inputs, on this account, play a role in nudging things one way or another, constraining what we see, but expectation or prediction play an equal, or even greater, role. The computer model gives us a concrete way of seeing how a process like this might actually work, which supports Føllesdal’s reading, and in turn provides a way of interpreting this model phenomenologically. Moreover, because we have implemented models and multiple studies of these models in relation to perceptual processes, the same kind of elaborative process discussed with edge detection—where nuances of the phenomenology are discovered thanks to ongoing research with computational models—could occur here as well.

A third example illustrates how these points apply to phenomenologists besides Husserl. Dreyfus’ account of skill is based on the work of Heidegger and Merleau-Ponty.⁴⁵ Skills and “coping” are nonconceptual and nonlinguistic.⁴⁶ You have to *show* someone how to tie their shoes, and then they need to practice doing it; you can’t just *tell* them what to do.⁴⁷ Words are inadequate in this domain. But what, then, to make of these nonconceptual, nonlinguistic coping skills? How to get a grip on them in a way that makes tractable

connections to brain and behavior? It turns out, computational techniques such as Fourier analysis, singular value decomposition (SVD), and detrended fluctuation analysis (DFA) are well-suited to deal with the problem. These techniques can be used to show how successful performance of a motor skill can be associated with specific sensorimotor patterns—narrow “bands” in high-dimensional action and coordination spaces—even when these patterns cannot easily be put into words.⁴⁸ Analysis of real-time behavioral data using these tools is increasingly common in psychology, and some have begun to apply these methods to the phenomenology of skill and engagement. For example, Heidegger’s famous distinction between “ready-to-hand” and “present-at-hand” modes of being (roughly, skillfully using things vs taking explicit notice of them) has been studied using DFA, in a video game with a controller that breaks halfway through an experimental trial. Results suggest that ready-to-hand use of entities is associated with “interaction dominant systems” with a specific complexity profile, and that cases of “breakdown” are associated with changes in complexity as measured by DFA. Ongoing work is carefully distinguishing behavioral competence from engagement in such tasks. So ideas introduced by Heidegger and Merleau-Ponty are being made precise using such tools as SVD and DFA. These quantities are extremely difficult to compute by hand, so computational models play a central role. Moreover, these computational models are publicly available and shareable across research groups, facilitating ongoing inquiry.⁴⁹

A final example illustrates computation-first computational phenomenology. Deep networks are neural networks with many processing layers. They have become the industry standard in visual pattern recognition tasks and have also become standard as scientific models of neural responses in the visual system.⁵⁰ Each layer in these models detects combinations of features from prior levels. For example, early layers detect edges (as noted above), and subsequent layers detect combinations of edges into textures, combinations of these textures into shapes, combinations of these shapes into more complex structures, and so on. In the 2010s, techniques emerged for visualizing these layers and their preferred inputs (that is, the inputs that cause them to be maximally active), as shown in Figure 1.⁵¹ The results are strange, but compelling. The images in the bottom panels feel plausible as part of what we pick up on when we see the images in the top panels. Just as we see edges, textures and surfaces, perhaps we also see more abstract structures like these, which are hard to characterize in words, but just seem like “something in all these things.” If so, this is a case of computationally-motivated discovery in phenomenology, a case where the computer model shows us something about the phenomenology that we would not otherwise have access to.

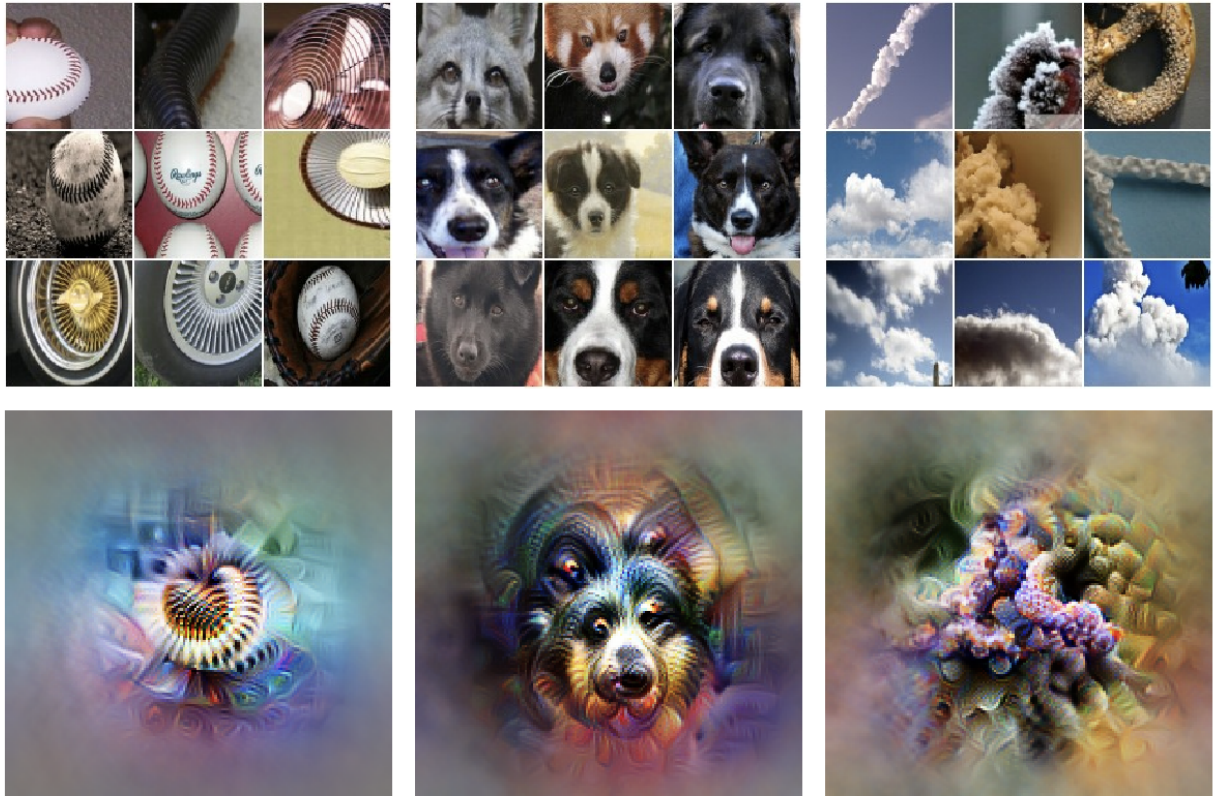


Figure 1. Preferred inputs for units in an internal processing layer of a deep convolutional network (bottom), and some images that cause a strong response in these units (top). From Olah, Mordvintsev, and Schubert (2017).

These four examples were chosen because they illustrate how phenomenological insights are possible through contact with computational modeling, and because they span several spatial and temporal scales in a way that is suggestive of future pluralistic integration. But there is a great deal of work in computational phenomenology beyond this.⁵² Perhaps the most prominent omissions in this discussion are computational studies of Husserl's theory of time consciousness and horizons, which have been examined by multiple authors.⁵³ There is also work in the phenomenological tradition that has not yet been linked to computational cognitive science, and in computational cognitive science that has not been linked to phenomenology. An important example on the computation-first side is the question of how transformers in large language models work. These are, after all, working models of linguistic competence, which arguably pass the Turing Test, and there are fascinating ideas emerging in the field of mechanistic interpretability to explain their inner workings, which may have relevance to phenomenology.⁵⁴ Recent work suggests that representations of words and word forms are processed, enhanced, and permuted as they move through the layers of a large language model in a way that might

shed light on hidden depths of the phenomenology of linguistic production. The account is fully oriented around ideas related to neural networks and distributed representations in neural networks but still has elements that are resonant with symbolic approaches and the traditional CTM. Thus, LLMs may suggest an approach to mental representation that straddles the old “classical/connectionist” divide in a nuanced way that sheds light on the equally subtle way our “predicative experience” of stable things emerges out of the more nuanced flux of passive synthesis.⁵⁵

Conclusion

By starting with the “given,” or “being,” and simply describing the world as they saw it, phenomenologists discovered new and suggestive processes, such as Husserl’s retentions and protentions, active and passive synthesis, horizons, and sedimentations; Heidegger’s ready-to-hand, present-at-hand, and the ecstatic structure of temporality, and Merleau-Ponty’s optimal grip and chiasm. Often, these ideas were developed in suggestive, but vague, ways. With the emergence and widespread availability of computational modeling tools, it has become possible to test and explore these ideas in greater depth and in a more communal way than was previously possible. We can now take something like Husserl’s model of time consciousness, implement a model of its physical basis on a computer simulation, and actually see it unfolding in a running model, which can be stopped, inspected in detail, tweaked, and easily shared with others. We have seen that this kind of computational phenomenology promises innovation, sometimes by starting with what a phenomenologist said and making it precise, sometimes by starting with something in a computational model and asking how it relates to our phenomenology. By using a convergent and communal approach, the hope is that plausible theories survive, and implausible theories are discarded over time.

The long-term goal of this type of project is a rich, integrated, account of consciousness and its relationship to its physical implementation. To speak imaginatively, the goal is to “feel” how conscious processes arise from or are instantiated in computational processes that occur in the brain. How much can the flow of information in a computer model of the brain reflect active and passive synthesis of streaming experiential data in an evolving field of consciousness? Can we develop a feeling for how the moment-to-moment experience of seeing a visual scene emerges from the flow of information through a hierarchy of feature detectors in the brain? Ideally, a coherent computational phenomenology will emerge in fits and starts from a patchwork of such accounts, some more symbolic, some based on IIT or active inference, and some based on bespoke models of specific processes. The hope is to facilitate this work by leveraging the virtues of computational modeling—the many insights it affords, the deep access it provides, and the fast agile community of researchers it supports.⁵⁶

Further Readings

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Williford, Kenneth. "A Brief on Husserl and Bayesian Perceptual Updating." *Axiomathes* 27, no. 5 (2017): 503-19.

Yoshimi, Jeff. "Phenomenology and Connectionism." *Frontiers in Psychology* 2 (2011): 288.

Yoshimi, Jeff. "Pluralist Neurophenomenology: A Reply to Lopes." *Phenomenology and the Cognitive Sciences* 24, no. 4 (2025): 897-920.

¹ One issue here is that transcendental phenomenology is supposed to abstract from external considerations about, for example, the brain or details of computational models, whereas everyday introspection does not preclude such considerations. However, one can arguably consider the results of everyday introspection independently of such external considerations, and then "import" those results into transcendental phenomenology. In fact, Husserl suggests that something like this is possible, when he allows that the results of "inauthentic calculations" can be checked against "authentic intuition." This is discussed further in section 2.

² Shaun Gallagher and Jesper Brøsted Sørensen, "Experimenting with phenomenology," *Consciousness and cognition* 15, no. 1 (2006): 119-134. Also see Gallagher and Gallese's contribution to this volume.

³ This goes by various names, which are reviewed in Yoshimi, Jeff. "Prospects for a naturalized phenomenology." In *Philosophy of Mind and Phenomenology*, pp. 287-309. Routledge, 2015.

⁴ In this work, Husserl can be understood either as an analytic philosopher clarifying how mathematics and algorithms work, or as a nascent phenomenologist, studying, among other things, the *phenomenology of computation*, the way mathematical objects are constituted in streams of experience, in part by way of algorithmic procedures. Either way, Husserl ends up saying things highly relevant to contemporary computational phenomenology.

⁵ The ability to see a group of things as a specific number of things is called "subitizing" in psychology. It has a long history, and there are many decades of accumulated results on the topic, concerning such things as proposed component mechanisms (in particular, attention and pattern recognition) and subitization ranges in different modalities. See Katzin, Naama, Zahira Ziva Cohen, and Avishai Henik, "If it looks, sounds, or feels like subitizing, is it subitizing? A modulated definition of subitizing," *Psychonomic Bulletin & Review* 26, no. 3 (2019): 790-797.

⁶ See Edmund Husserl, *Philosophy of Arithmetic: Psychological and Logical Investigations with Supplementary Texts from 1887-1901*, transl. Dallas Willard (Dordrecht: Kluwer Academic, 2003), xviii.

⁷ Husserl, 2003, xxxix.

⁸ Husserl, 2003, 138.

⁹ Husserl, 2003, 138.

¹⁰ Husserl, 2003, 202.

¹¹ Husserl, 2003, 139. Husserl distinguishes between calculation (*Rechnung*), algorithms (*Algorithmen*) and machines (*Maschinen*). The distinction seems to be that calculations are operations using rules for replacing signs with signs, algorithms are broader systems combining multiple calculation procedures into a system of rules, and machines, in relation to these, are implementations of calculations and algorithms (that is, “computations”) that unfold in accordance with causal laws.

¹² Edmund Husserl, *Hua Mat I*, 314, quoted in Carlo Ierna, “Husserl on Mechanical Calculation: Outsourcing Symbolic Intentionality,” in *Husserl and the Awakened Reason: Critical Essays on Mathematics, Science and Phenomenology*, eds. Iulian Apostolescu and Carlo Ierna (2026).

¹³ This aspect of Husserl is developed at length in Philipp Berghofer. *The Justificatory Force of Experiences*. Synthese Library 427. Cham: Springer, 2022.

¹⁴ Ierna (2026).

¹⁵ See Dreyfus, Hubert, and Harrison Hall, eds. *Husserl, Intentionality, and Cognitive Science* Cambridge, MA: MIT Press, 1982.

(1984); Ronald McIntyre, “Husserl and the representational theory of mind,” *Topoi* 5, no. 2 (1986): 101–113. For a critical discussion of this idea, see Beth Preston, “Husserl’s non-representational theory of mind,” *The Southern journal of philosophy* 32, no. 2 (1994): 209.

¹⁶ James M. Edie, “Husserl’s conception of ‘the grammatical’ and contemporary linguistics,” in *Readings on Edmund Husserl’s logical investigations*, eds. J. N. Mohanty and Robert Sokolowski (Dordrecht: Springer Netherlands, 1977), 137–61.

¹⁷ Dreyfus and Hall (1984) and Robert M. Harlan, “Towards a computational phenomenology,” *Man and World* 17, no. 3 (1984): 261–277.

¹⁸ Dieter Münch, “The Early Work of Husserl and Artificial Intelligence,” *Journal of the British Society for Phenomenology* 21, no. 2 (1990): 107–120.

¹⁹ Harlan (1984), 272.

²⁰ Harlan (1984), 270.

²¹ See Jesse Lopes, *Computational foundations of phenomenology* (PhD diss., Boston University, 2020); Jesse Lopes, “How do mental processes preserve truth? Husserl’s discovery of the computational theory of mind,” *Husserl Studies* 36, no. 1 (2020): 25–45; Jesse Lopes, “Phenomenology as Proto-Computationalism: Do the Prolegomena Indicate a Computational Reading of the Logical Investigations?” *Husserl Studies* 39, no. 1 (2023): 47–68.

²² Edmund Husserl, *Logical Investigations*, trans. J. N. Findlay, 2 vols. (London: Routledge & Kegan Paul, 1970), 50.

²³ Husserl, Edmund (1970), 50.

²⁴ On connectionism and dynamical systems, see Jeff Yoshimi, “Phenomenology and connectionism.” *Frontiers in psychology* 2 (2011): 288; on dynamical systems, see Tim van Gelder. “Wooden Iron? Husserlian Phenomenology Meets Cognitive Science.” *Psyche* 2 (1996); on Bayesian approaches, see Kenneth Williford, “A brief on Husserl and Bayesian perceptual updating,” *Axiomathes* 27, no. 5 (2017): 503–519.

²⁵ See Jeff Yoshimi. “Pluralist neurophenomenology: A reply to Lopes.” *Phenomenology and the Cognitive Sciences* (2023).

²⁶ Larissa Albantakis, Leonardo Barbosa, Graham Findlay, Matteo Grasso, Andrew M. Haun, William Marshall, and Giulio Tononi. (2023). Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLoS computational biology*, 19(10), e1011465.

²⁷ At least in terms of what is discussed or cited in published accounts. William James, who probably should be classified as a phenomenologist, is discussed. It is canonical figures such as Husserl or Merleau-Ponty who are not discussed.

²⁸ Sergio Ponce de Leon and Jeff Yoshimi, in preparation. Also see Marek Pokropski, “Phenomenology and mechanisms of consciousness: Considering the theoretical integration of phenomenology with a

mechanistic framework," *Theory & Psychology* 29, no. 5 (2019): 601–619; and Ishan Singhal, Ramya Mudumba, and Narayanan Srinivasan, "In search of lost time: Integrated information theory needs constraints from temporal phenomenology," *Philosophy and the Mind Sciences* 3 (2022), 1–19.

²⁹ Adenauer G. Casali, Olivia Gosseries, Mario Rosanova, Mélanie Boly, Simone Sarasso, Karina R. Casali, Silvia Casarotto, Marie-Aurélie Bruno, Steven Laureys, Giulio Tononi, and Marcello Massimini, "A theoretically based index of consciousness independent of sensory processing and behavior," *Science translational medicine* 5, no. 198 (2013): 198ra105–198ra105.

³⁰ On the experience of space, see Andrew Haun and Giulio Tononi, "Why does space feel the way it does? Towards a principled account of spatial experience," *Entropy* 21, no. 12 (2019): 1160. On the experience of time, see Renzo Comolatti, Matteo Grasso, and Giulio Tononi, "Why does time feel the way it does? Towards a principled account of temporal experience," arXiv:2412.13198 (2024).

³¹ On the PyPhi package, see William GP Mayner, William Marshall, Larissa Albantakis, Graham Findlay, Robert Marchman, and Giulio Tononi, "PyPhi: A toolbox for integrated information theory," *PLoS computational biology* 14, no. 7 (2018): e1006343.

³² See Maxwell JD, Ramstead, Anil K. Seth, Casper Hesp, Lars Sandved-Smith, Jonas Mago, Michael Lifshitz, Giuseppe Pagnoni et al, "From generative models to generative passages: A computational approach to (neuro) phenomenology" *Review of Philosophy and Psychology* 13, no. 4 (2022): 829–857.

³³ In personal communication, Maxwell Ramstead offered the following helpful clarification on these points: "The term 'predictive processing' is an umbrella or family resemblance term ... which denotes a family of theories with loosely similar theoretical commitments: Bayesian brain, predictive coding, active inference, FEP, and so on. ... One salient difference between predictive coding and active inference is what the generative model is. In predictive coding, the generative model is an internal model encoded by the brain. In active inference, the generative model describes the whole agent–environment system. The internal model is known as the variational density."

³⁴ Mahault Albarracin, Riddhi J. Pitliya, Maxwell JD Ramstead, and Jeff Yoshimi, "Mapping Husserlian phenomenology onto active inference", In *International workshop on active inference*, (Cham: Springer Nature Switzerland, 2022), 101.

³⁵ An overview discussion is given in Thomas Parr, Giovanni Pezzulo, and Karl J. Friston, *Active inference: the free energy principle in mind, brain, and behavior*, (Cambridge, MA: MIT Press, 2022). For a discussion of the field of psychiatry and clinical psychology, see Paul B. Badcock and Christopher G. Davey, "Active Inference in Psychology and Psychiatry: Progress to Date?," *Entropy* 26, no. 10 (2024): 833.

³⁶ Ramstead et al., From generative models to generative passages, 829.

³⁷ Ramstead et al., From generative models to generative passages, 829.

³⁸ Ramstead et al., From generative models to generative passages, 831.

³⁹ See Albarracin et al, Mapping Husserlian phenomenology onto active inference., 99–111.

⁴⁰ "Spatial scale" in phenomenology can refer to something like location in a containment hierarchy: sensory experience is a part of perceptual experience and so it is at a lower spatial scale. See Yoshimi, (2023).

⁴¹ Edmund Husserl, *Analyses Concerning Passive and Active Synthesis: Lectures on Transcendental Logic*, trans. Anthony J. Steinbock (Dordrecht: Kluwer Academic Publishers, 2001), 194.

⁴² For a historical survey and contemporary review, which includes review of the relevant neuroscience, see Qinyuan Huang, and Jiaxiong Huang, "Comprehensive review of edge and contour detection: From traditional methods to recent advances," *Neural Computing and Applications* 37, no. 4 (2025): 2175–2209.

⁴³ Aron Gurwitsch, *The collected works of Aron Gurwitsch (1901–1973): vol. III: The field of consciousness: Theme, thematic field, and margin*, (Dordrecht: Springer, 2010), 262–63.

⁴⁴ Føllesdal, Dagfinn "Husserl's Theory of Perception," in Dreyfus and Hall (1982).

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- ⁴⁵ Hubert L. Dreyfus, "A Phenomenology of Skill Acquisition as the Basis for a Merleau-Pontian Nonrepresentational Cognitive Science." Unpublished manuscript, 2002. Available at PhilArchive.
- ⁴⁶ Hubert L. Dreyfus, "Overcoming the myth of the mental: How philosophers can profit from the phenomenology of everyday expertise," *Proceedings and addresses of the American Philosophical Association*, 79, no. 2, (2005), 47-65.
- ⁴⁷ Hubert L. Dreyfus and Stuart E. Dreyfus, "What artificial experts can and cannot do," *AI & Society* 6, no. 1 (1992), 18-26.
- ⁴⁸ Tim Meyer, Arnold D. Kim, Michael Spivey, and Jeff Yoshimi, "Mouse tracking performance: A new approach to analyzing continuous mouse tracking data," *Behavior Research Methods* 56, no. 5 (2024), 4682-94.
- ⁴⁹ The original analysis of ready-to-hand and breakdown using DFA is Dobomir Dotov, Lin Nie, and Anthony Chemero, "A demonstration of the transition from ready-to-hand to unready-to-hand," *PloS One* 5, no. 3 (2010): e9433. The follow-up described is currently underway but is partially developed in Timothy Scott Meyer, *The complexity of engagement*, PhD diss., University of California, Merced, 2022.
- ⁵⁰ Daniel LK Yamins, Ha Hong, Charles F. Cadieu, Ethan A. Solomon, Darren Seibert, and James J. DiCarlo, "Performance-optimized hierarchical models predict neural responses in higher visual cortex," *Proceedings of the national academy of sciences* 111, no. 23 (2014), 8619-24. [I will note that after publication of this chapter, this highly relevant paper by Keisuke Suzuki came to my attention: <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2026.1819038/full>]
- ⁵¹ Chris Olah, Alexander Mordvintsev, and Ludwig Schubert, "Feature Visualization," *Distill* 2, no. 11 (2017): e7. <https://doi.org/10.23915/distill.00007>.
- ⁵² Much of this work is referenced in other notes above. Ramstead et al. (2022), contains further references. On psychopathology see Aaron L. Mishara and Philipp Sterzer, "Phenomenology is Bayesian in its application to delusions," *World Psychiatry* 14, no. 2 (2015), 185.
- ⁵³ A recent discussion of computational theories of time consciousness, which critically surveys some of the earlier work is Juan Diego Bogotá, "Can the predictive mind represent time? A critical evaluation of predictive processing attempts to address Husserlian time-consciousness," *Phenomenology and the Cognitive Sciences* (2023), 1-21. The relationship between horizons and early AI systems are discussed (both positively and critically) by Dreyfus and Hall (1982) and Harlan (1984). A more recent discussion of the computational basis of horizons is in Yoshimi (2011).
- ⁵⁴ See Pierre Beckmann and Matthieu Queloz, "Mechanistic Indicators of Understanding in Large Language Models," arXiv:2507.08017 (2025).
- ⁵⁵ See Yoshimi (2023) and Jeff Yoshimi, "Husserl's theory of belief and the Heideggerian critique," *Husserl Studies* 25, no. 2 (2009), 121-40.
- ⁵⁶ I am grateful to Ken Archer, Pierre Beckmann, Mark Ettinger, Mirja Hartimo, Johannes Kleiner, Andrew Lee, Maxwell Ramstead, and Sergio Ponce de Leon for helpful discussion and feedback.